DAISY: Dual-Granularity Contrastive Learning for Disease Similarity Prediction

Zihao Gao, Huifang Ma, Yike Wang, Zhixin Li, and Liang Chang

***Abstract*—The quantification of similarities among human diseases is crucial for enhancing our understanding of disease biology, which can aid in improving disease diagnosis, prognosis and treatment, and drug development. Recently, efforts have been devoted to quantifying disease similarity by integrating multi-view data sources from disparate biological data. However, disease data are often sparse due to their rarity or privacy, leading to suboptimal representation of disease when biological entity relationships and labeled disease data are not adequately modeled. To address this challenge, we propose DAISY (Dual-grAnularity contrastIve learning for diSease similaritY prediction), an effective deep learning model for disease representation modeling. DAISY leverages a novel dual-granularity contrastive learning mechanism to enhance the representation of diverse biological entities. Its ability to enable the collaborative supervision of diseases represented by both homogeneous and heterogeneous information is of great significance, leading to a high level of disease representation learning. Besides, this contrastive learning mechanism combines two independent and complementary components: a hierarchical biological entity relationship-aware module to capture disease features on other biological entities, and a disease association capturing module based on signed random walk with precious disease data. Experimental results demonstrate that DAISY achieves outstanding performance on the disease similarity prediction problem.**

***Index Terms*—Disease similarity prediction, disease representation learning, contrastive learning, signed graph.**

I. Introduction

THE measurement of the similarity of human diseases is a thriving field in modern disease biology. Researchers attempt to quantify the similarities between different diseases across various biological contexts. Recent studies have confirmed the existence of genetic and molecular similarities among diseases with dissimilar phenotypic traits [1]. For instance, Fig. 1(a) illustrates four diseases with distinct clinical symptoms that share genetic variants [2], while Fig. 1(b) reveals that the association between HIV and SARS-CoV-2 has led to the repurposing of azvudine, originally used to treat AIDS, to combat COVID-19 [3]. Simultaneously, the development of high-throughput molecular analysis technologies has enabled data-driven approaches to advance our understanding of disease relationships systematically. Therefore, utilizing available biomedical data to design accurate and effective disease similarity measures is of paramount importance and great value for understanding pathogenesis, improving disease diagnosis, prognosis, and treatment, as well as drug repositioning.

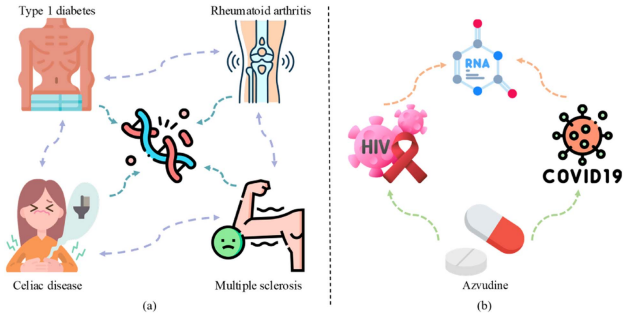


Fig. 1. (a) Patients with type 1 diabetes exhibit sudden weight loss, patients with rheumatoid arthritis exhibit joint swelling, patients with celiac disease exhibit diarrhea, and patients with multiple sclerosis exhibit limb weakness. These four diseases with distinct clinical symptoms may share genetic variants [2]; (b) Azvudine is an RNA-related polymerase inhibitor, which inhibits the reverse transcription process of HIV by inhibiting RNA-related polymerase. By discovering the similar characteristics of SARS-CoV-2 and HIV in the process of replication, researchers have decided to use Azvudine, which is used to treat AIDS, to treat COVID-19 [3].

Up to now, there are three main categories of methods for measuring disease similarity: molecular feature-based, phenotype-based, and semantic-based. Molecular-based methods rely on qualitative associations between molecules from relevant data sources (e.g., disease-associated genes, proteins, non-coding RNAs) and diseases to assess similarity by calculating the overlap of two disease-associated molecular sets [4], [5]. Phenotype-based methods use the explicit diagnosis given by the physician of the patient's symptoms to measure the disease [6]. Semantic-based methods utilize the graph structure of disease-related ontologies, such as biomedical texts, to quantify disease similarity

Received 8 July 2023; revised 19 February 2024, 25 May 2024, and 15 January 2025; accepted 23 January 2025. Date of publication 30 January 2025; date of current version 3 April 2025. This work was supported in part by the Gansu Province Key Foundation Project under Grant 24JRRA123, in part by the Industrial Support Project of Gansu Colleges, China, under Grant 2022CYZC11, and in part by the National Natural Science Foundation of China under Grant 62441701, Grant 62276073, and Grant U22A2099. *(Corresponding author: Huifang Ma.)*

Zihao Gao and Yike Wang are with the College of Computer Science and Engineering, Northwest Normal University, Lanzhou 730070, China.

Huifang Ma is with the College of Computer Science and Engineering, Northwest Normal University, Lanzhou 730070, China, and also with the Guangxi Key Laboratory of Trusted Software, Guilin University of Electronic Technology, Guilin 541004, China (e-mail: mahuifang@yeah.net).

Zhixin Li is with the School of Computer Science and Engineering, Guangxi Normal University, Guilin 541004, China.

Liang Chang is with the School of Computer Science and Information Security, Guilin University of Electronic Technology, Guilin 541004, China.

The implementation of DAISY is available at https://github.com/Jcmorz/DAISY.

This article has supplementary downloadable material available at https://doi.org/10.1109/TCBBIO.2025.3536100, provided by the authors.

Digital Object Identifier 10.1109/TCBBIO.2025.3536100

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by integrating the information content of the most informative common ancestor for the pairwise diseases [7], [8], [9]. However, these non-parametric models have limitations such as inefficiency and lacking objectivity. Moreover, diseases in these approaches are represented by discrete or sparse feature vectors that fail to capture the deep semantic information of diseases.

Inspired by the success of graph deep learning in bioinformatics, Li et al. obtained disease representations by reconstructing a comprehensive gene network using a robust and flexible network embedding model with combinations of disease-associated genes [10]. Chen et al. employed the Graph Convolutional Networks (GCNs) model to learn gene network and Gene Ontology (GO) graph [11]. They refined gene representations by means of cross-view contrastive learning and also followed Li et al.'s approach to obtain disease representations. Despite these significant advances in studying disease similarity, there are still two crucial issues that need to be addressed.

1. *How to acquire precise semantic information about diseases?* Several methodologies have been exploited to capture the connotation of disease similarity by fusing as much disease-related biological knowledge as possible. However, many of these methods rely on weighting feature dimensions with different coefficients, which may lead to subjective assessments of disease similarity. To accurately capture the deep semantic information of diseases, it is important to flexibly encode disease-related biological knowledge to refine disease attribute features, rather than simply fusing as much knowledge as possible. This will help to avoid subjective assessments of disease similarity and ensure more precise acquisition of semantic information.

2. *How to effectively integrate the precious supervisory information in disease similarity tasks?* Tackling sparse supervisory information can be a significant challenge in disease similarity prediction tasks. Many disease similarity prediction methods rely heavily on molecular, ontological, and semantic information to calculate disease similarity. However, these methods often do not make reasonable use of sparse and valuable disease supervisory information. To address this, it is critical to efficiently encode the disease supervisory signal into a universal basis intrinsic to the graph, without relying on any disease-related biological data. This will help to better incorporate sparse and precious disease supervisory information, leading to more accurate disease similarity calculations.

To address the aforementioned challenges, we propose a novel deep learning model called DAISY, which stands for Dual-grAnularity contrastIve learning for diSease similaritY prediction. *To tackle the first challenge,* we design a hierarchical biological entity relationship-aware module to capture the characteristics of diseases in relation to other biological entities. This module consists of three levels. At the first level, the gene ontologies are modeled using relation-sensitive GCNs, and the fine-grained gene representations are generated using gene ontologies. At the second level, gene graph is augmented using Singular Value Decomposition (SVD), and the fine-grained contrastive learning mechanism is performed on the two gene graphs, thus refining the gene representations. At the third level, disease representations are obtained through combinations of disease-related genes. *To address the second challenge,* we elaborately model precious disease data as a signed graph and perform signed random walk with modified balance theory. Our method allows for more discriminative disease representations, which can effectively differentiate between similar and dissimilar diseases. Finally, coarse-grained contrastive learning is employed to enable diseases represented by homogeneous and heterogeneous information to collaboratively supervise each other. This dual-granularity contrastive learning mechanism plays a key role in enhancing the model's ability to distinguish subtle differences and patterns in data. By clearly comparing positive samples with negative samples, the model can learn more fine-grained and coarse-grained distinguishing features, thus improving its ability to generalize disease representation in various situations, and ultimately leading to more accurate disease similarity prediction.

In summary, we have made the following major contributions in this work:

- We have designed a hierarchical biological entity relationship-aware module to accurately capture the deep semantic information of diseases. Unlike traditional methods that rely solely on coefficient-weighted values for different dimensions to determine disease similarity, our module captures the features of diseases in relation to various biological entities.
- We have explicitly modeled the disease data, which has received little scrutiny in previous studies. Our proposed disease association capturing module is based on signed random walk, enabling the generated disease representation to distinguish between similar and dissimilar diseases more effectively.
- We have designed a novel dual-granularity contrastive learning mechanism to enhance the representation of different biological entities. Its ability to promote collaborative supervision of diseases represented by homogeneous and heterogeneous information shows great promise for improving disease representation and advancing our understanding of complex biological systems.
- We have conducted extensive experiments on a real-world dataset to demonstrate the effectiveness of our proposed model in comparison to classic and state-of-the-art methods.

II. Materials and Preliminaries

A. Datasets

- *GO graph*: The GO knowledgebase is the world's largest source of information on the functions of genes [12], [13]. GO annotations are divided into three top-level branch terms: biological_process (BP), molecular_function (MF), and cellular_component (CC). All GO terms and the set of inclusion relationships form a hierarchy and directed acyclic graph of GO. This GO graph can be modeled as a multi-relational graph in our framework. Unlike other disease similarity methods, we use all three branches of GO instead of only considering the BP branch.
- *GO-Gene associations*: The GO-gene associations are downloaded from the gene database of the National Center for Biotechnology Information (NCBI) to bridge gene instances and ontology concepts.

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- *Gene interaction network*: The gene interaction network utilized in this paper is derived from HumanNet [14], which encompasses 99.8% of Homo sapiens' protein-encoding genes. HumanNet is constructed by integrating expanded data using network inference algorithms. The edges in this network are assigned weights calculated through Bayesian statistics algorithms, specifically employing a log-likelihood score. HumanNet supports a three-tier model comprising the protein–protein physical interaction network (HumanNet-PI), the functional gene network (HumanNet-FN), and the functional network extended by co-citation (HumanNet-XC). For our analysis, we employ HumanNet-FN as the training data.
- *Gene-diaease associations*: The gene-disease associations are obtained from the DisGeNET database [15]. DisGeNET serves as a comprehensive knowledge management platform that effectively integrates and standardizes data concerning disease-associated genes and variations. This powerful platform encompasses an extensive range of information, encompassing over 24,000 diseases, 17,000 genes, and 117,000 genomics variations.

B. Notation

We denote vectors in lowercase boldface, matrices in uppercase boldface, sets in uppercase calligraphic font, and scalars in normal character. For a vector $\mathbf{x}$, $[\mathbf{x}]_i$ denotes the ith element of $\mathbf{x}$. For a matrix $[\mathbf{X}]$, $[\mathbf{X}]_i$ denotes the vector on its ith row, $[\mathbf{X}]_{ij}$ denotes the (i, j)th element of $[\mathbf{X}]$. The superscript $(\cdot)^\top$ denotes the matrix transpose. For a set $\mathcal{X}$, $|\mathcal{X}|$ denotes the number of elements contained in $\mathcal{X}$.

C. Definition

Definition II.1 (GO graph). Gene ontology (GO) knowledgebase is the source of information on the functions of genes. All GO terms and the set of inclusion relationships form a hierarchy and directed acyclic graph of GO. Thus, let a directed and labeled multi-relational graph $\mathcal{G}^o = (\mathcal{V}^o, \mathcal{E}^o, \mathcal{R}^o)$ denote GO graph, where $\mathcal{V}^o$ is GO term set and $\mathcal{E}^o$ is edge set. $(v_i^o, r^o, v_j^o) \in \mathcal{E}^o$, where $r^o \in \mathcal{R}^o$ is a relation type. To make the GO graph more suitable for our model processing, we add inverse relation and self-connection to $\mathcal{R}^o$.

Definition II.2 (Gene graph). Let an undirected weighted graph $\mathcal{G}^g = (\mathcal{V}^g, \mathcal{E}^g, \mathcal{W}^g)$ denote gene graph, where $\mathcal{V}^g$ is gene set, $\mathcal{E}^g$ is the edge set, in which each edge connects two genes with the same function, $\mathcal{W}^g$ is the set of weights of all edges, where each weight w_{ij}^g of edge (v_i^g, v_j^g) is calculated as normalized log-likelihood score based on the Bayesian statistics algorithm.

Definition II.3 (GO-gene graph & gene-disease graph). The GO-gene graph can be denoted as an undirected graph $\mathcal{G}^{og} = (\mathcal{V}^{og}, \mathcal{E}^{og})$, where $\mathcal{V}^{og} = \mathcal{V}^o \cup \mathcal{V}^g$ represents the set of nodes that correspond to GO terms and genes, $\mathcal{E}^{og}$ denotes the set of edges and indicates the existence of interaction between two kinds of nodes. Similarly, the gene-disease graph can be denoted as an undirected graph $\mathcal{G}^{gd} = (\mathcal{V}^{gd}, \mathcal{E}^{gd})$.

Definition II.4 (Disease signed graph). As disease similarity data comprises both positive and negative samples, it is natural to model it as a signed graph $\mathcal{G}^d = (\mathcal{V}^d, \mathcal{E}^+, \mathcal{E}^-)$, where $\mathcal{V}^d$ is disease set while $\mathcal{E}^+ \subseteq \mathcal{V}^d \times \mathcal{V}^d$ and $\mathcal{E}^- \subseteq \mathcal{V}^d \times \mathcal{V}^d$ denote the sets of positive and negative edges, respectively, where $\mathcal{E}^+ \cap \mathcal{E}^- = \emptyset$.

Definition II.5 (Disease similarity prediction). Disease similarity prediction is a computational task in biomedical informatics that aims to quantify similarities between diseases based on various data sources. The task involves developing algorithms to analyze and integrate multiple data sources to generate disease similarity scores or rankings. In this paper, our goal is to learn a mapping function $f((v_i^d, v_j^d)|\theta) : \hat{y}_{ij} \to [0, 1]$ from binary groups to scores, where θ denotes the model parameters, $\hat{y}_{ij}$ denotes the prediction similarity score between v_i^d and v_j^d.

III. The Proposed DAISY Model

Fig. 2 illustrates our proposed model termed DAISY. For one thing, we adopt a hierarchical biological entity relationship-aware module to flexibly encode biological knowledge related to diseases. This effectively refines the attribute characteristics of diseases and accurately captures the deep semantic information of diseases. For another, we model disease data as a signed graph and perform random walk on the $\mathcal{G}^d$ to generate disease representations that can finely differentiate between similar and dissimilar diseases. Finally, we have developed a dual-granularity contrastive learning mechanism to enhance the representation of biological entities across different stages. The contrastive learning of disease makes the diseases represented by homogeneous information and heterogeneous information collaborate with each other and high-level disease representations are thus learned for similarity prediction.

A. Hierarchical Biological Entity Relationship-Aware Module

In this subsection, we introduce our hierarchical biological entity relationship-aware module, which is designed to extract the attribute characteristics of diseases by flexibly encoding the biological knowledge related to diseases. We begin by noting that genomics has significantly contributed to a growing collection of gene-function and gene-disease annotations that can be exploited by informatics to study the similarity between diseases. Such an approach can yield valuable insight into disease etiology and reveal common pathophysiology. However, estimating disease similarity solely on the basis of shared genes can be misleading, as various combinations of genes may be associated with similar diseases, especially for complex diseases. To overcome this limitation, it is necessary to identify common biological processes instead of only explicit gene matches between diseases [4]. Therefore, at the first level, we employ GO processes to enhance simple gene information, so as to generate fine-grained gene representation.

The main hurdle in defining a convolution operation on $\mathcal{G}^o$ is the handling of different types of edges. To address this, we draw inspiration from the Relational Graph Convolutional Networks presented in [16] and introduce a relation-sensitive GCNs. The propagation process of the GO term $[\mathbf{X}^o]_i$ on the $\mathcal{G}^o$ is defined

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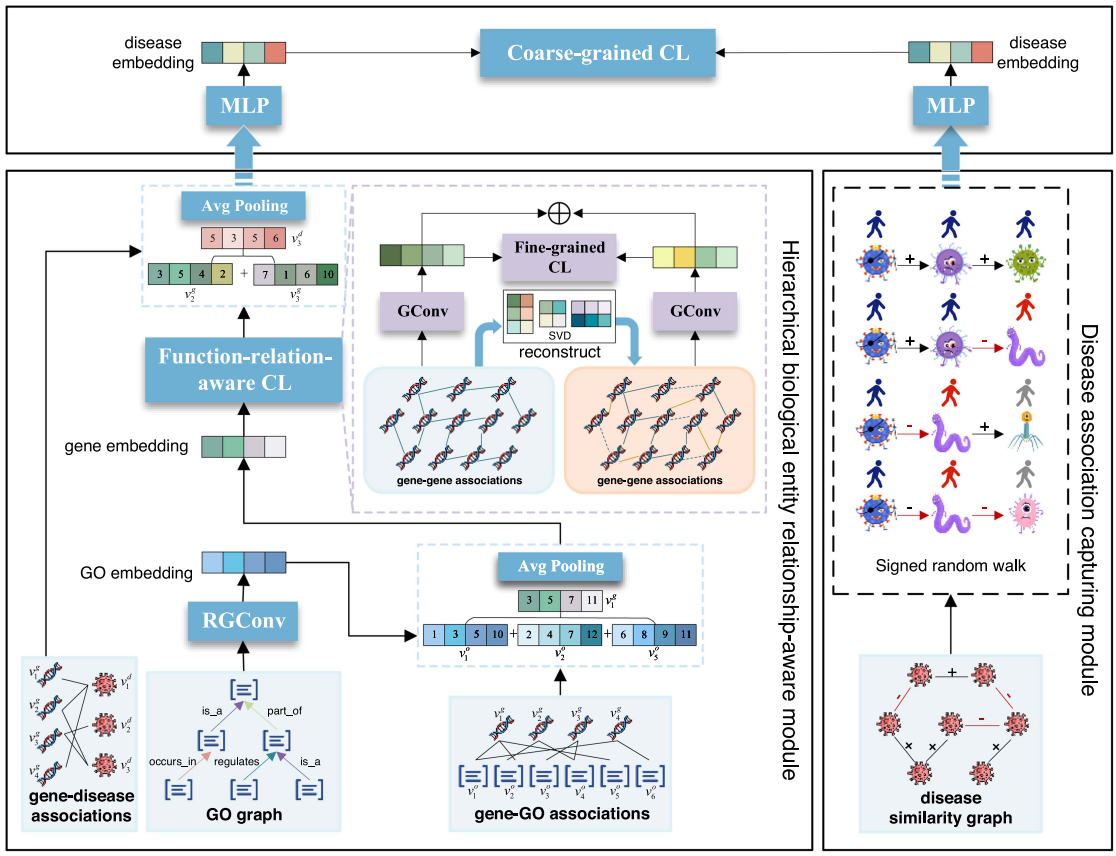


Fig. 2. Illustration of the proposed DAISY, which is formed by organically combining two core components by a dual-granularity contrastive learning mechanism. (1) Hierarchical biological entity relationship-aware module uses relation-sensitive GCNs to extract structural information and semantic relations from gene ontologies and generates fine-grained gene representation. Further, it uses GCNs to model genes from gene function graph, and ultimately, disease representation $\mathbf{x}_1^d$ is fused by related gene representations. (2) Disease association capturing module employs the modified balance theory to perform a signed random walk on the elaborately modeled signed disease association graph, so as to generate a discriminatory disease representation $\mathbf{x}_2^d$. (3) Dual-granularity contrastive learning mechanism is designed to enhance the representation of different biological entities. The mechanism is powerful for promoting collaborative supervision of diseases represented by homogeneous and heterogeneous information, leading to high-level disease representation $\mathbf{x}^d$.

as:

$$[\mathbf{X}^o]_i^{(l+1)} = \sum_{r^o \in \mathcal{R}^o} \sum_{j \in \mathcal{N}_i^{r^o}} \frac{1}{|\mathcal{N}_i^{r^o}|} [\mathbf{W}_{r^o}]^{(l)} [\mathbf{X}^o]_j^{(l)} + [\mathbf{W}_c]^{(l)} [\mathbf{X}^o]_i^{(l)}, \quad (1)$$

where $\mathcal{N}_i^{r^o}$ denotes the set of neighbor indices of GO term o_i under relation $r^o \in \mathcal{R}^o$. $[\mathbf{W}_{r^o}]^{(l)}$ is the relation-specific weight matrix in the lth layer and $[\mathbf{W}_c]^{(l)}$ is the weight of self-connection in the lth layer. Note that, we use one-hot coding as the initial feature of GO terms.

Then, we perform average pooling on $\mathcal{G}^{og}$ to obtain the representation of each gene:

$$[\mathbf{X}^g]_i^{(0)} = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} [\mathbf{X}^o]_j, \quad (2)$$

where $[\mathbf{X}^g]_i^{(0)}$ indicates the initial representation of gene g_i obtained by the pooling of GO term representations and $\mathcal{N}_i$ denotes the set of neighbor indices of gene v_i^g on $\mathcal{G}^{og}$.

At the second level, we aim to encode the gene embedding on undirected weighted gene graph $\mathcal{G}^g$ which represents the relationships between genes. The incidence matrix $[\mathbf{A}^g] \in \mathbb{R}^{|\mathcal{V}^g| \times |\mathcal{V}^g|}$ is defined based on the gene set $\mathcal{V}^g$, where $[\mathbf{A}^g]_{ij} = w_{ij}^g$ if there is an edge between genes v_i^g and v_j^g, and $[\mathbf{A}^g]_{ij} = 0$ otherwise. To further refine the incidence matrix, we define $[\widehat{\mathbf{A}}^g] = [\mathbf{A}^g] + [\mathbf{I}]$, where $[\mathbf{I}]$ is the identity matrix. Additionally, we use the diagonal degree matrix $\widehat{\mathbf{D}} \in \mathbb{R}^{|\mathcal{V}^g| \times |\mathcal{V}^g|}$ to weigh $\mathcal{G}^g$. Finally, we apply graph convolution to extract meaningful features and relationships between genes as follows:

$$[\mathbf{X}^g]^{(l+1)} = [\widehat{\mathbf{D}}]^{-1} [\widehat{\mathbf{A}}^g] [\mathbf{X}^g]^{(l)} [\mathbf{W}_g]^{(l)}, \quad (3)$$

where $[\mathbf{W}_g]^{(l)}$ is the weight matrix to do feature transformation in the lth layer. In each convolution, the genes gather information from their neighbors. By doing so, the learned $[\mathbf{X}^g]$ can capture the relevant genomic information.

The third level of our hierarchical biological entity relationship-aware module aims at obtaining disease representations through combinations of disease-related genes. The resulting disease representations can absorb rich semantics of ontology structure and gene function, and capture the complex relationships between genes and diseases as:

$$[\mathbf{X}_1^d]_i = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} [\mathbf{X}^g]_j, \quad (4)$$

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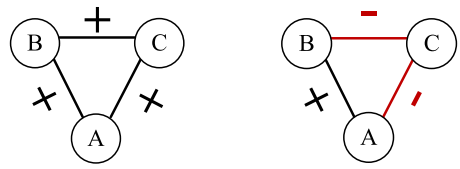


Fig. 3. Relationship combination of triads in structural balance theory.

where $[\mathbf{X}_1^d]_i$ is the representation of disease v_i^d obtained by the pooling of gene representations and $\mathcal{N}_i$ denotes the set of neighbor indices of disease v_i^d on $\mathcal{G}^{gd}$.

B. Disease Association Capturing Module

In this subsection, we introduce the innovative disease association capturing module. Our disease supervision data consists of positive and negative samples, and as such, we model this valuable information as a signed graph elaborately. This graph can be considered a special case of heterogeneous graph, consisting of edge annotations to indicate either positive edge (similar) or negative edge (dissimilar). To obtain more discriminating disease representations, i.e., the message propagation from similar/dissimilar diseases to a target disease should be distinguished, we perform a signed random walk on $\mathcal{G}^d$. However, the traditional signed random walk is based on the simple structural balance theory, which focuses on the balance judgment of triangles [17]. Fig. 3 illustrates the fundamental principles of structural balance theory, encompassing four intuitive conclusions: 1) Friend's friend is friend; 2) Friend's enemy is enemy; 3) Enemy's friend is enemy; 4) Enemy's enemy is friend. This theory fails to account for intricate relationships involving consecutive edges, such as friend-of-enemy or enemy-of-enemy. To this end, we introduce a modified balance theory, which incorporates a sign into the random surfer, enabling it to consider negative edges [18]. As a result, we can directly and efficiently encode the disease supervision signal into the underlying universal base of the graph with local structure context.

Let $[\mathbf{A}^d]$ be the signed adjacency matrix of $\mathcal{G}^d$, where $[\mathbf{A}^d]_{ij}$ is positive or negative when there is a positive or a negative edge from disease v_i^d to disease v_j^d respectively, and zero otherwise. $[\mathbf{D}^d]$ is the out-degree diagonal matrix. Semi-row normalized matrix is $[\widetilde{\mathbf{A}}^d] = [\mathbf{D}^d]^{-1}[\mathbf{A}^d]$. Then, positive and negative semi-row normalized matrix $[\widetilde{\mathbf{A}}_+^d]$ and $[\widetilde{\mathbf{A}}_-^d]$ which contains only positive and negative values can be defined as:

$$[\widetilde{\mathbf{A}}_+^d] = \max([\widetilde{\mathbf{A}}^d], \mathbf{0}), \tag{5}$$

$$[\widetilde{\mathbf{A}}_-^d] = \min([\widetilde{\mathbf{A}}^d], \mathbf{0}), \tag{6}$$

where $\max(\cdot,\cdot)$ compares two matrices and returns a new matrix containing the element-wise maxima, and $\min(\cdot,\cdot)$ compares two matrices and returns a new matrix containing the element-wise minima.

Thus, the process of Signed Random Walk with Restart (SRWR) can be defined as:

$$[\mathbf{X}_+^d]^{(l+1)} = (1-\alpha)([\widetilde{\mathbf{A}}_+^d]^\top[\mathbf{X}_+^d]^{(l)} + \beta[\widetilde{\mathbf{A}}_-^d]^\top[\mathbf{X}_-^d]^{(l)} + (1-\gamma)[\widetilde{\mathbf{A}}_+^d]^\top[\mathbf{X}_-^d]^{(l)}) + \alpha\mathbf{p}, \tag{7}$$

$$[\mathbf{X}_-^d]^{(l+1)} = (1-\alpha)([\widetilde{\mathbf{A}}_-^d]^\top[\mathbf{X}_+^d]^{(l)} + \gamma[\widetilde{\mathbf{A}}_+^d]^\top[\mathbf{X}_-^d]^{(l)} + (1-\beta)[\widetilde{\mathbf{A}}_-^d]^\top[\mathbf{X}_-^d]^{(l)}), \tag{8}$$

$$[\mathbf{X}^d]^{(l+1)} = [\mathbf{X}_+^d]_i^{(l+1)} - [\mathbf{X}_-^d]^{(l+1)}, \tag{9}$$

where $[\mathbf{X}_+^d]^{(l)}$ represents the trust SRWR score vector for all nodes in the lth layer. $[\mathbf{X}_-^d]^{(l)}$ corresponds to the distrust SRWR score vector for all nodes in the lth layer. The parameter α signifies the restart probability, determining the likelihood of restarting the random walk process. β denotes that if the negative surfer at node m encounters a negative edge, its sign becomes positive with probability or negative with $1-\beta$ at node t. Similarly, parameter γ functions in a similar manner to β. $\mathbf{p}$ is a vector where only the sth element is set to 1, and all other elements are 0, representing a specific node or position. Note that the restarting vector $\mathbf{p}$ is not added to $[\mathbf{X}_-^d]^{(l+1)}$ in this case because the surfer's sign must become positive when it goes back to the seed node.

After the random walk on $\mathcal{G}^d$ has converged, we denote the representation of the disease v_i^d as $[\mathbf{X}_2^d]_i$.

C. Prediction Generation and Model Optimization

Thus far, we have acquired two types of disease representations that are modeled from distinct perspectives: disease representation $[\mathbf{X}_1^d]_i$, which contains rich semantics through the encoding of biological knowledge related to diseases, and disease representation $[\mathbf{X}_2^d]_i$, which equips discrimination by employing a modified balance theory for performing signed random walk.

Given a disease v_i^d, we add up two representations of the disease for the final similarity prediction:

$$[\mathbf{X}^d]_i = [\mathbf{X}_1^d]_i + [\mathbf{X}_2^d]_i, \tag{10}$$

Therefore, the similarity between two diseases can be measured by cosine similarity:

$$\hat{y}_{ij} = \frac{[\mathbf{X}^d]_i \cdot [\mathbf{X}^d]_j}{||[\mathbf{X}^d]_i|| \times ||[\mathbf{X}^d]_j||}, \tag{11}$$

where $||\cdot||$ denotes L2-norm.

We formulate the learning objective as a cross-entropy loss function:

$$\mathcal{L}_p = -\frac{1}{|\mathcal{T}|}\sum_{(i,j)\in\mathcal{T}} y_{ij}\log\hat{y}_{ij} + (1-y_{ij})\log(1-\hat{y}_{ij}), \tag{12}$$

where $\mathcal{T}$ is the sampled training set of disease pairs and y_{ij} denotes true label between disease v_i^d and v_j^d.

D. Enhancing DAISY With Dual-Granularity Contrastive Learning

Inspired by the recent success of contrast learning in dealing with highly sparse data, we design a dual-granularity contrast learning mechanism, which contains two sub-modules, namely, fine-grained contrast learning and coarse-grained contrast learning. In fine-grained contrastive learning, we measure the similarity of diseases by analyzing the information at the gene level. Gene is the genetic material encoding protein in organism, and

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the function of protein is closely related to the development of diseases. By comparing the differences of genetic variation such as gene expression and mutation between diseases, the fine-grained contrastive learning module can effectively compare the subtle differences between diseases. The coarse-grained contrastive learning module focuses on similarity measurement at the disease level. The disease is the manifestation of human dysfunction, which involves many aspects, including symptoms, pathological process, clinical manifestations and so on. By analyzing the characteristics and correlation of diseases, we can reveal their commonness and similarity. The coarse-grained contrastive learning module can comprehensively compare diseases. Combining fine-grained and coarse-grained contrastive learning modules, we can evaluate the similarity between diseases at different levels. This dual-granularity design can fully consider the biological characteristics of genes and diseases and provide comprehensive and accurate similarity measurement results.

To avoid noise interference caused by random graph augmentation and to preserve the intrinsic semantic structure of the $\mathcal{G}^g$ as much as possible, we introduce graph augmentation technology with SVD. While contrastive learning methods prove effective, state-of-the-art techniques in contrastive learning encounter several inherent limitations [19]. First, graph augmentation through random perturbation risks losing valuable structural information, thus misleading contrastive representation learning. Second, the efficacy of heuristic-guided representation contrasting schemes heavily relies on the quality of the view generator, constraining model generality and making it susceptible to noisy user behavior data. Third, many current GCNs-based contrastive learning methods face challenges due to over-smoothing, resulting in indistinguishable representations. However, our approach leverages SVD for contrastive augmentation, facilitating unconstrained structure refinement through global collaborative relation modeling. Specifically, we first perform SVD on the adjacency matrix $[\mathbf{A}^g]$, such that $[\mathbf{A}^g] = [\mathbf{U}][\mathbf{S}][\mathbf{V}]^\top$, where $[\mathbf{U}]$ and $[\mathbf{V}]$ are orthonormal matrix with columns being the eigenvectors of $[\mathbf{A}^g]$'s row-row / column-column correlation matrix. $[\mathbf{S}]$ is diagonal matrix storing the singular values of $[\mathbf{A}^g]$, and the largest singular values are usually associated with the principal components of the matrix. We then truncate the list of singular values to keep only the largest q values and augment the adjacency matrix with the truncated matrices as follows:

$$[\mathbf{A}^g_q] = [\mathbf{U}_q][\mathbf{S}_q][\mathbf{V}_q]^\top, \tag{13}$$

where $[\mathbf{U}_q]$ and $[\mathbf{V}_q]$ contain the first q columns of $[\mathbf{U}]$ and $[\mathbf{V}]$ respectively. $[\mathbf{S}_q]$ is the diagonal matrix of the q largest singular values.

Then, according to (3), two embeddings $[\mathbf{X}^g_1]^{(l+1)}$ and $[\mathbf{X}^g_2]^{(l+1)}$ of each gene from the original gene graph and the augmented gene graph can be obtained. Further, we utilize NT-Xent [20] as our objective function to enable the two representations of gene to collaboratively supervise each other:

$$\mathcal{L}_g = -\frac{1}{|\mathcal{V}^g|}\sum_{i=1}^{|\mathcal{V}^g|}\log\frac{e^{sim([\mathbf{X}^g_1]_i,[\mathbf{X}^g_2]_i)/\tau}}{\sum_{j=1}^{|\mathcal{V}^g|}e^{sim([\mathbf{X}^g_1]_i,[\mathbf{X}^g_2]_j)/\tau}}, \tag{14}$$

where τ denotes the temperature parameter and $sim(\cdot,\cdot)$ is the cosine similarity.

The gene representations in (4) can be obtained using the sum of two representations of genes:

$$[\mathbf{X}^g]_i = [\mathbf{X}^g_1]_i + [\mathbf{X}^g_2]_i, \tag{15}$$

Similarly, we utilize contrastive learning to refine the disease representations, so that diseases represented by homogeneous and heterogeneous information can collaborate and supervise each other:

$$\mathcal{L}_d = -\frac{1}{|\mathcal{V}^d|}\sum_{i=1}^{|\mathcal{V}^d|}\log\frac{e^{sim([\mathbf{X}^d_1]_i,[\mathbf{X}^d_2]_i)/\tau}}{\sum_{j=1}^{|\mathcal{V}^d|}e^{sim([\mathbf{X}^d_1]_i,[\mathbf{X}^d_2]_j)/\tau}}, \tag{16}$$

Finally, the total loss $\mathcal{L}$ is defined as:

$$\mathcal{L} = \mathcal{L}_p + \lambda_1\mathcal{L}_g + \lambda_2\mathcal{L}_d, \tag{17}$$

where λ_1 and λ_2 are hyperparameters to control the scale of contrastive learning. The complete algorithm is summarized in Algorithm 1.

E. Model Generalization

After several rounds of parameter optimization, the output of the DAISY gradually converges. When dealing with unseen diseases, the initial step involves querying the DisGeNET database to retrieve genes associated with the unseen disease. Subsequently, the trained gene embeddings are utilized to obtain embeddings for the unseen disease. Finally, by computing the cosine similarity between the embeddings of the unseen disease and known diseases, we can quantify the similarity between the two diseases. As researchers identify an increasing number of disease associations through clinical trials, DAISY undergoes periodic retraining to update the embeddings of genes in the HumanNet. This iterative training process enhances the model's ability to predict the similarity between unseen and known diseases more effectively.

IV. Experiments

A. Experimental Settings

1) Benchmark: When evaluating predictive models, the selection of high-quality benchmarks as ground truth is crucial. To ensure consistency with other disease similarity methods, we adopt the same benchmark set as used in their studies [10], [21]. This benchmark set combines two manually examined datasets, both of which contain pairs of diseases exhibiting a high degree of similarity [5], [22]. The first dataset is initially released by [5], and subsequent literature verification conducted by [4] further identified and labeled similar disease pairs. The second dataset comprises disease pairs deemed similar by resident doctors, based on the analysis of electronic health records from the American population [22].

2) Basline: For comparison, we use the following competitive methods as baselines:

- *Resnik* [23]: Resnik's method employs the "is_a" relationship present in the Disease Ontology to calculate the similarity between pairs of diseases. By leveraging the

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Algorithm 1: Disease Similarity Prediction Procedure for DAISY.

Input: GO graph $\mathcal{G}^o$; gene graph $\mathcal{G}^g$; GO-gene graph $\mathcal{G}^{og}$; gene-disease graph $\mathcal{G}^{gd}$; disease signed graph $\mathcal{G}^d$; maximum training epochs T.

Output: Mapping function $f((v_i^d, v_j^d)|\theta) : \hat{y}_{ij} \rightarrow [0, 1]$ from binary groups to scores.

Initialize $\theta = \{[\mathbf{W}_r], [\mathbf{W}_c], [\mathbf{W}_g], [\mathbf{W}_a], [\mathbf{W}_m]\}$ with Xavier initialization;
Perform SVD on the adjacency matrix of the gene graph $\mathcal{G}^g$;
Augment the adjacency matrix with the truncated matrices via (13);
$t \leftarrow 1$;
repeat
  for $v \in \mathcal{V}^o$ **do**
    Calculate $[\mathbf{x}^o]$ using relation-sensitive GCNs for relational modeling via (1);
  end for
  Initialize $[\mathbf{x}^g]^{(0)}$ via (2);
  for $v \in \mathcal{V}^g$ **do**
    Calculate gene embedding $[\mathbf{x}_1^g]$ from gene graph $\mathcal{G}^g$ via (3);
    Calculate gene augmented embedding $[\mathbf{x}_2^g]$;
  end for
  Obtain the fused gene embedding $[\mathbf{x}^g]$ via (15);
  Obtain disease embedding $[\mathbf{x}_1^d]$ via (4);
  Perform signed random walk to obtain disease embedding $[\mathbf{x}_2^d]$ via (7)(8)(9);
  Obtain the fused disease embedding $[\mathbf{x}^d]$ via (10);
  Calculate the similarity score between any two diseases $\hat{y}_{ij}$ via (11);
  Evaluate main loss $\mathcal{L}_p$ via (12);
  Evaluate fine granularity contrastive loss $\mathcal{L}_g$ via (14);
  Evaluate coarse granularity contrastive loss $\mathcal{L}_d$ via (16);
  Update θ by optimizing (17);
  $t \leftarrow t + 1$;
until $t > T$ or (17) is converged;
return $f((v_i^d, v_j^d)|\theta)$;

hierarchical structure of the Disease Ontology, Resnik's method quantifies the extent of similarity between diseases based on their shared ancestry.

- *Lin* [7]: Lin has proposed the information theory definition of similarity, and a method to evaluate the semantic similarity of diseases by considering the commonness and differences between terms.
- *Wang* [24]: Wang's approach involves encoding the semantic information of GO terms into numerical values, which is achieved by aggregating the semantic contributions of the ancestor terms in the GO graph. By considering multiple common ancestors and using two types of relationships in a mixed measure, Wang then calculates the semantic similarity between GO terms.
- *PSB* [4]: PSB computes disease similarity based on biological process GO terms, which consider information from both co-occurrences in annotations and ontological hierarchy.
- *SemFunSim* [21]: SemFunSim leverages both the functional similarity of gene interactions obtained from the human network and the semantic similarity between disease ontologies. It quantifies disease similarity by considering the associated genes and the common ancestor nodes of the two diseases.
- *Li* [10]: Li's method incorporates both a gene interaction network and GO hierarchy to reconstruct a novel gene network. Subsequently, the method computes disease similarity by measuring the cosine similarity between disease embeddings, which are derived from the relevant gene representations.
- *CoGO* [11]: CoGO integrates the knowledge of gene interaction network and GO terms by graph deep learning model, and then maximizes the consistency of corresponding gene-GO associations by contrastive learning. Disease representation is obtained by pooling related gene representations, and the similarity between diseases is inferred by cosine similarity.

3) Evaluation Metrics: Evaluation metric plays a critical role in achieving the optimal discriminator in the inference stage. Following [11], [25], we use Area Under Receiver Operating Characteristic (AUROC) and Area Under Precision-Recall curve (AUPR) to evaluate the prediction results. We run experiments five times with different random seeds and report their average performance.

4) Implementation Details: We implement our DAISY with PyTorch [26] and adopt Adam [27] optimizer to train the proposed DAISY. For the general setting, the embedding size is 64, the L_2 regularization is $1e-8$, the initial learning rate is 0.0001, the training epochs are set to 200, and the temperature τ is set as 2. For the hyperparameters in the disease association capturing module, the restart probability α is 0.1, and the balance parameters β and γ are 0.5 and 0.7, respectively. Besides, the total number of model parameters is 107,077,125 and the model size is 408.48 MB. For the baseline models, we refer to their best parameter setups reported in the original papers and directly report their results if available. Particularly, for Li's method, we report its result from CoGO's original paper since we use the same datasets and evaluation settings.

B. Experimental Results

1) Overall Performance: We obtain the final evaluation results through the evaluation index of five repeated experiments on average. In addition, we deploy the pairwise t-test under both metrics. Table I reports the performance comparison results. Fig. 4 shows the ROC curve, PR curve and corresponding AUROC, AUPRC obtained by our method and other two deep learning methods. We have the following observations:

Obs1: The efficacy of methods based on deep learning consistently surpasses that of traditional feature design methods. For benchmark datasets, it becomes evident that deep learning-based

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TABLE I
THE AVERAGE PERFORMANCE OF THE COMPARED METHODS OBTAINED BY 5 REPLICATE EXPERIMENTS

	Methods	Indicators AUROC	AUPR
Feature Design Methods	Resnik	0.6022	-
	Lin	0.5890	-
	Wang	0.6133	-
	PSB	0.6467	-
	SemFunSim	0.6814	-
Deep Learning Methods	Li	0.7818	0.5776
	CoGO	0.8317	0.7733
	DAISY	**0.9566**	**0.7748**

The best results (according to the pairwise *t*-test with 95% confidence) are in bold.

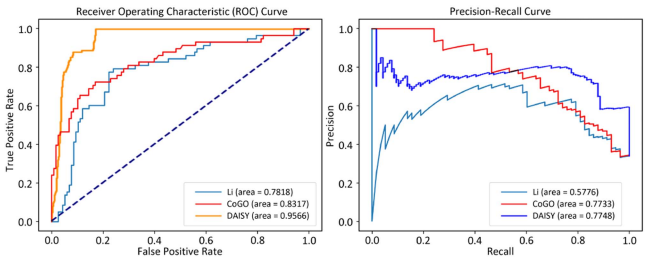


Fig. 4. The ROC curve, PR curve and corresponding AUROC, AUPRC obtained by our method and other two deep learning methods.

methods, such as Li's method and CoGO, consistently outperform traditional feature design methods. This compelling evidence underscores the inadequacy of simplistic disease features designed solely based on the overlap between disease-related gene sets or the hierarchical distance in the ontology. These conventional approaches fail to encapsulate the intricate and abstract features inherent in disease data.

Obs2: DAISY has emerged as the best performance among all the comparison methods. While CoGO introduces a more potent feature mining technique compared to Li's method, it overlooks the utilization of crucial disease supervision information. On the other hand, although Li's method incorporates valuable supervision information, its implementation fails to leverage it effectively, resulting in even inferior performance than CoGO. Besides, we compare the training time of three deep learning models. Although Li's model adopts advanced graph learning model, the model training time is 102 minutes on Intel Core i7-13650HX CPU @ 4.90 GHz. In contrast, COGO can complete training in 17 minutes on the same equipment. The training and disease similarity reasoning of DAISY takes 20 minutes, which is close to that of CoGO model. In light of these limitations, the proposed DAISY framework not only exhibits a rational approach to modeling disease features but also meticulously designs a dual-granularity contrastive learning mechanism. This strategic design aims to harness the full potential of both knowledge information and supervision data, thereby maximizing their impact on performance outcomes.

2) Generalization Analysis of DAISY: To ensure an unbiased evaluation of the results obtained from the cross-validation set, we conducted independent testing on all deep learning models for an objective assessment. All models were trained on the cross-validation set and then tested on a separate and independent test set. This independent testing protocol helps evaluate the performance of DAISY on unseen data, as the model parameters and training processes are unrelated to the independent test set. The experimental results have been summarized in Table II. The results from independent testing align with those from cross-validation, indicating that DAISY performs well not only on known data but also generalizes effectively to unseen data, which is crucial for ensuring the practicality and reliability of the model.

TABLE II
THE PERFORMANCE OF THE DEEP LEARNING METHODS EVALUATED ON THE INDEPENDENT TEST SETS

Methods	Indicators AUROC	AUPR
Li	0.8083	0.5981
CoGO	0.8524	**0.7957**
DAISY	**0.9631**	0.7892

Furthermore, it is observed that in terms of AUROC, the performance of CoGO is significantly inferior to that of DAISY, both in cross-validation and independent testing. However, in independent testing, CoGO slightly outperforms DAISY in terms of AUPR. In this scenario, CoGO tends to classify more samples as positive, resulting in a higher number of true-positive samples compared to DAISY. Nevertheless, the significantly lower AUROC of CoGO compared to DAISY suggests that CoGO predicts a much higher number of false-positive samples than DAISY. When the model predicts a large number of false positive samples, doctors may base their diagnoses and treatment plans on incorrect similarity predictions. If the two diseases are not genuinely similar but are erroneously classified as such by the model, doctors may misdiagnose or mistreat patients, thereby delaying proper treatment.

3) Ablation Study on DAISY: To verify the effectiveness of the hierarchical biological entity relationship-aware module and disease association capturing module, we introduce the following three variants of DAISY:

- DAISY-w/o DAC: DAISY without the disease association capturing module.
- DAISY-w/o HER: DAISY without hierarchical biological entity relationship-aware module.
- DAISY-w/o DCL: DAISY without dual-granularity contrastive learning. Note that, the augmented gene graph is also abandoned.

We show the performance of DAISY and its three variants in Fig. 5. We can observe that by removing any of the modules, the performance of the model shows different magnitudes of degradation. Specifically, DAISY with disease association capturing module shows better performance than any other variant. Furthermore, compared with DAISY, DAISY-w/o DAC shows a significant performance decline, indicating that the disease association capturing module plays a more important role in enhancing the representation of diseases to be discriminative. Meanwhile, the performance of DAISY-w/o HER shows promising performance. However, we argue that merely performing

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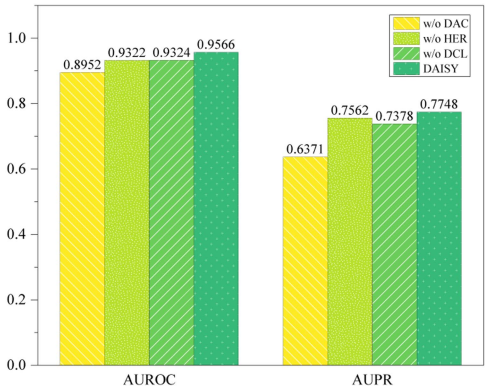


Fig. 5. The comparison of different components in DAISY.

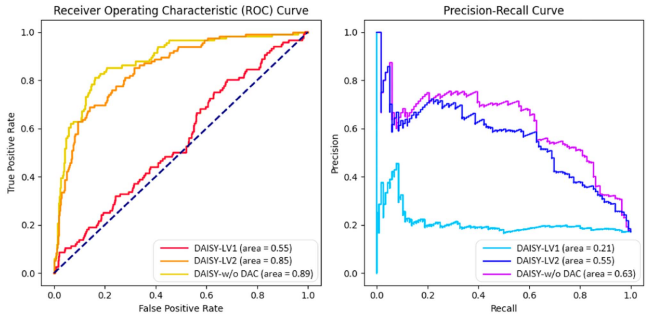


Fig. 6. ROC curve, PR curve and according to AUROC scores and AUPR scores from three hierarchical biological entity relationship-aware module variants.

TABLE III
THE PERFORMANCE OF VARIANT DAISY-CRPT COMPARED WITH DAISY

Methods	Indicators	
	AUROC	AUPR
DAISY-CRPT	0.9527	0.7707
DAISY	**0.9566**	**0.7748**

random walk exhibits limitations for disease similarity prediction tasks, primarily due to their lack of interpretability. Moreover, disregarding disease-related data hampers the model's ability to generalize effectively to novel disease scenarios.

To our profound astonishment, the removal of the dual-granularity contrastive learning from our model results in a decline in performance. DAISY-w/o DCL is not even as good as that predicted by single view under AUPR metric. This outcome substantiates the assertion that simple fusion of disease representations generated by two independent modules fails to yield improved results. Instead, it detrimentally affects the quality of representations generated by each individual module. In contrast, the dual-granularity contrastive learning mechanism we introduced successfully orchestrates the harmonious integration of these two independent modules, thereby improving the representation of biological entities across varying levels. This mechanism assumes importance as it facilitates the collaborative supervision capability of diseases represented by both homogeneous and heterogeneous information, ultimately culminating in high-level disease representation learning.

4) Ablation Study on Hierarchical Biological Entity Relationship-Aware Module: To thoroughly evaluate the effectiveness of each level in our hierarchical biological entity relationship-aware module, we further design the following two variants and analyze them together with DAISY-w/o DAC:

- DAISY-LV1: We initialize the gene representation with vectors of all zeros, and then update the gene representations on the gene graph $\mathcal{G}^g$ constructed by HumanNet according to the description of (3), through the graph convolution network. The regenerated gene representations are averaged and pooled by using the gene-disease graph $\mathcal{G}^{gd}$ as described in (4), thus generating disease representations.
- DAISY-LV2: Based on DAISY-w/o DAC, we further remove the fine-grained contrastive learning module. That is, first, on the GO graph $\mathcal{G}^o$, according to the description of (1), the GO term representations are updated through the relation-sensitive GCNs. The regenerated GO term representations are average-pooled by using GO-gene graph $\mathcal{G}^{og}$ as described in (2), thus generating gene representations. Then, the gene representations continue to follow the flow of DAISY-LV1 to further generate disease representations.

In Fig. 6, we show the ROC curves and PR curves of the three models. Intuitively, DAISY-w/o DAC performs the best, followed by DAISY-LV2, and DAISY-LV1 is relatively poor. By comparing DAISY-LV2 and DAISY-LV1, we find that it is difficult to fully learn the key causes of diseases only by gene associations. However, when we introduce GO term information into the model, the performance of the model has been significantly improved, which shows that GO term information carries rich semantic information of pathogenic genes.

In addition, the application of fine-grained contrastive learning refines the gene representations obtained from GO terms, which enables the model to generate more advanced disease embedding. Therefore, we conclude that combining the GO term information with fine-grained contrastive learning can not only improve the performance of the model, but also better capture the complex relationship between diseases and genes.

5) Effectiveness Analysis of Constructing Contrastive View by SVD: To verify the effectiveness of constructing contrastive view by SVD in fine-grained contrastive learning, we follow Velickovic et al. [28], and develop a variant DAISY-CRPT. This DAISY-CRPT uses the approach of random corruption to create another view. Specifically, after using GConv to obtain gene embedding matrix $[\mathbf{X}^g]$ on the $\mathcal{G}^g$, we shuffle $[\mathbf{X}^g]$, and the new matrix obtained is used as gene embedding matrix in the contrastive view. The experimental results are shown in Table III. It can be observed that the performance of DAISY-CRPT is slightly worse than that of data augmentation based on SVD, which may be due to the random corruption that cannot identify the important structure in the $\mathcal{G}_g$. On the contrary, the enhanced view constructed by SVD can retain the key graph structure, which is very important for acquiring disease embedding with accurate gene semantics.

6) Hyperparameter Sensitivity Analysis of Disease Association Capturing Module: There are three key hyperparameters in the disease association capturing module, i.e., the restart

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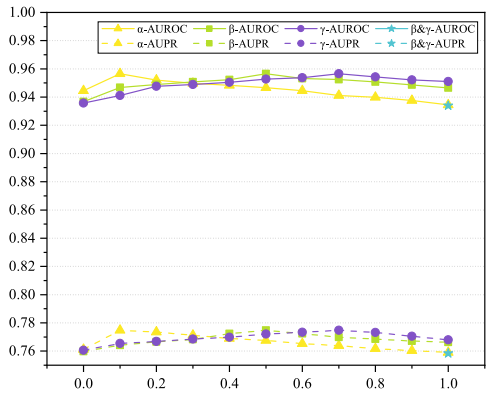


Fig. 7. Impact of three hyperparameters in disease association capturing module on model performance.

probability α and the balance parameters β and γ. To further investigate the impact of these three hyperparameters on the performance of DAISY, we range these three hyperparameters within {0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1}. Additionally, when studying a specific parameter, the other two parameters are kept at their optimal values. Fig. 7 illustrates the impact of different values for these three parameters on the performance of DAISY. Solid lines represent the results under the AUROC evaluation, while dashed lines represent the results under the AUPR evaluation. From the figure, it is observed that the model performance reaches its optimum when the restart probability α is set to 0.1. As the restart probability increases, the performance gradually decreases. This decrease in performance could be attributed to the model's limited ability to explore new states within the state space. With a higher restart probability, the model is more likely to abandon previously acquired useful knowledge and restart the exploration process from a random state. This behavior can hinder the model's ability to converge to optimal solutions or explore promising areas in the state space. Moreover, larger values for β and γ achieved high performance in this experiment. Specifically, the model performance reached its peak when β and γ were set to 0.5 and 0.7, respectively. Finally, we have conducted a separate test by setting both β and γ to 1. It is observed that the ideal balance theory does not apply to real-world disease similarity prediction tasks, as both AUROC and AUPR metrics are compromised when $\beta = 1$ and $\gamma = 1$.

7) Hyperparameter Sensitivity Analyzation of Dual-Granularity Contrastive Learning: Our method relies on two crucial parameters, λ_1 and λ_2, which respectively control the weights of fine-grained and coarse-grained contrastive learning. To evaluate the sensitivity of these parameters, we perform a grid search over a range $0 \leq \lambda_1, \lambda_2 \leq 1$ by 0.2 (i.e., search (λ_1, λ_2) in $\mathbf{S} = \{(0.1x, 0.1y) | 0 \leq x, y \leq 10 \& x, y \in \mathbb{Z}\}$). Noting that $\lambda_1 = 0$ is equivalent to removing fine-grained contrastive learning, while $\lambda_2 = 0$ is equivalent to eliminating coarse-grained contrastive learning. As illustrated in Fig. 8, our experiments reveal that DAISY has achieved the best performance when λ_2 is set between 0.4 and 0.8. In terms of λ_1, we can observe that when λ_1 ranges between 0.6 and 1, DAISY demonstrates the best performance in AUROC. This suggests

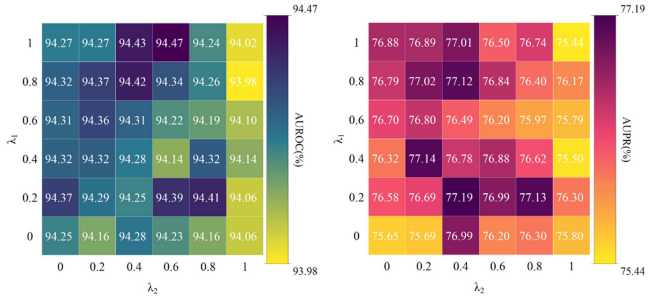


Fig. 8. Hyperparameter sensitivity of loss coefficients λ_1 and λ_2.

that fine-grained contrastive learning plays a crucial role in dual-granularity contrastive learning, particularly in identifying false positive samples. On the other hand, when λ_1 is set between 0.2 and 0.4, DAISY exhibits the best performance in AUPR, indicating that coarse-grained contrastive learning is more effective in identifying false negative samples. It is worth noting that when λ_1 is set to 0.5, DAISY considers the contrastive learning of both granularities, achieving optimal values in both AUROC and AUPR. However, since the increment of the scale line of Fig. 8 is 0.2, this observation cannot be directly discerned from the figure. Based on these findings, we set λ_1 and λ_2 to 0.5 ultimately.

V. Related Work

A. Contrastive Learning

Contrastive learning has achieved significant success in computer vision [29], [30], [31] and natural language processing tasks [32], [33] by maximizing the consistency between different views. In recent years, contrastive learning has also garnered attention in the field of Graph Neural Networks (GNNs) [20]. Within the realm of GNNs, contrastive learning methods can be applied at both the node level [34], [35] and the graph level [28], [36]. Previously, researchers have utilized random graph structure or embedding corruption as a form of data augmentation for contrastive learning [37], [38]. While this method can boost model efficacy, it falls short in terms of interpretability. Furthermore, the random corruption approach may inadvertently discard vital information, possibly worsening issues related to data sparsity. In response to these challenges, subsequent research has turned to heuristic-based approaches for creating diverse views in contrastive learning [39], [40]. Our study integrates a heuristic strategy with more reliable data augmentation techniques to realize dual-granularity contrastive learning. This approach not only enhances the precision of embeddings but also preserves the integrity of key information, leading to more interpretable predictions in disease similarity.

B. Disease Similarity Prediction

Early studies on disease similarity prediction have focused on network analysis [4], [7], [21], [23], and have studied disease similarity by analyzing the connection mode between nodes, such as path similarity and module discovery. For instance, Ni et al. [41] measure disease associations by integrating

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disease-gene associations with protein-protein interaction networks. Oerton et al. [42] combine six types of biological data to construct disease networks, employing similarity fusion to predict disease relationships. With the advancement of deep learning, researchers are applying these techniques to uncover the relationships between genetic factors and disease phenotypes. For example, Li et al. [10] develop network embeddings to capture gene semantics, while MTGNN [25] introduces multitask optimization strategies that enhance disease similarity prediction accuracy through few-shot learning. Contrastive learning offers inherent advantages for similarity prediction by constructing positive and negative sample pairs to measure similarities directly. In object detection, assessing the similarity between different image features facilitates more robust matching [31], [43]. However, due to the irregularity and complexity of graph data, existing contrastive learning methods are often tailored for computer vision tasks and do not readily transfer to graph learning applications. CoGO [11] addressed this by refining gene embeddings through cross-view contrastive learning. Distinct from them, our work employs a dual-granularity contrastive learning method to directly refine disease embeddings, thereby achieving more accurate disease similarity prediction.

VI. Conclusion and Future Works

In this paper, we propose a novel deep learning model, called DAISY, for quantifying the similarity between human diseases. There are several problems we intend to pursue in our future research endeavors. First, the pathogenesis of a disease is a complex interplay of numerous factors. Presently, most methods focus on identifying the disease's origins solely from internal factors within the human body, neglecting the irreversible damage inflicted by the patient's environment. Second, we recognize that the current dataset for model training is limited in scale. This scarcity poses a hindrance to researchers interested in disease similarity prediction tasks, impeding further advancements in the field. To mitigate this challenge, we aspire to construct reliable and comprehensive datasets that encompass a broader spectrum of disease instances. Finally, we suggest particular attention be paid to the false-positive rate of the model when considering its application in real-world scenarios. By reducing the rate of false alarms, the risks faced by both patients and healthcare professionals can be mitigated, thereby further enhancing the reliability and practicality of the model.

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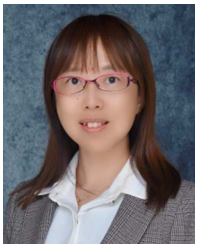
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Zihao Gao is currently working toward the master degree with the College of Computer Science and Engineering, Northwest Normal University. His main research interest focuses on artificial intelligence in drug discovery.



Huifang Ma received the BE degree from Northwest Normal University, China, in 2003, the MS degree from Beijing Normal University, China, in 2006, and the PhD degree from the Institute of Computing Technology, Chinese Academy of Sciences, in 2010. She is now a professor with the College of Computer Science and Engineering, Northwest Normal University. Her research interests include data mining and machine learning.



Yike Wang is currently working toward the master degree with the College of Computer Science and Engineering, Northwest Normal University. Her main research interest focuses on artificial intelligence-aided drug side effect prediction.



Zhixin Li received the BS and MS degrees from the Huazhong University of Science and Technology, in 1992 and 2004, respectively, and the PhD degree in computer software and theory from the Institute of Computing Technology, Chinese Academy of Sciences, in 2010. He is a professor with the School of Computer Science and Engineering, Guangxi Normal University, P.R. China. His research interests include image understanding, machine learning, and cross-media computing. He has won the best doctoral dissertation award of the Chinese Association of Artificial Intelligence in 2011.



Liang Chang is currently a professor with the Guilin University of Electronic Technology. His research interests include knowledge mapping, intelligent recommendation, dynamic system verification, data privacy protection and so on.

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